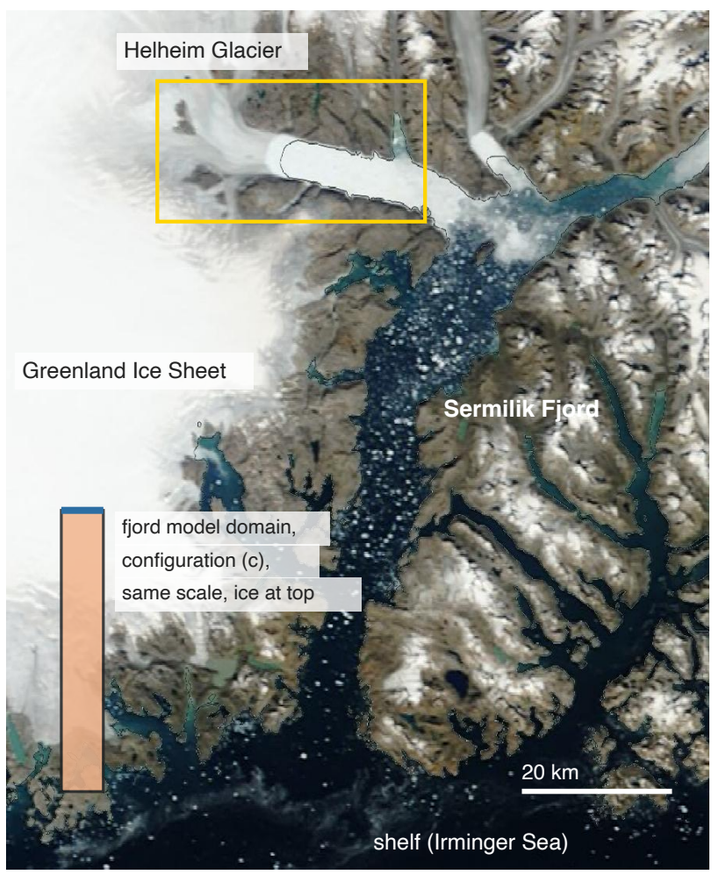
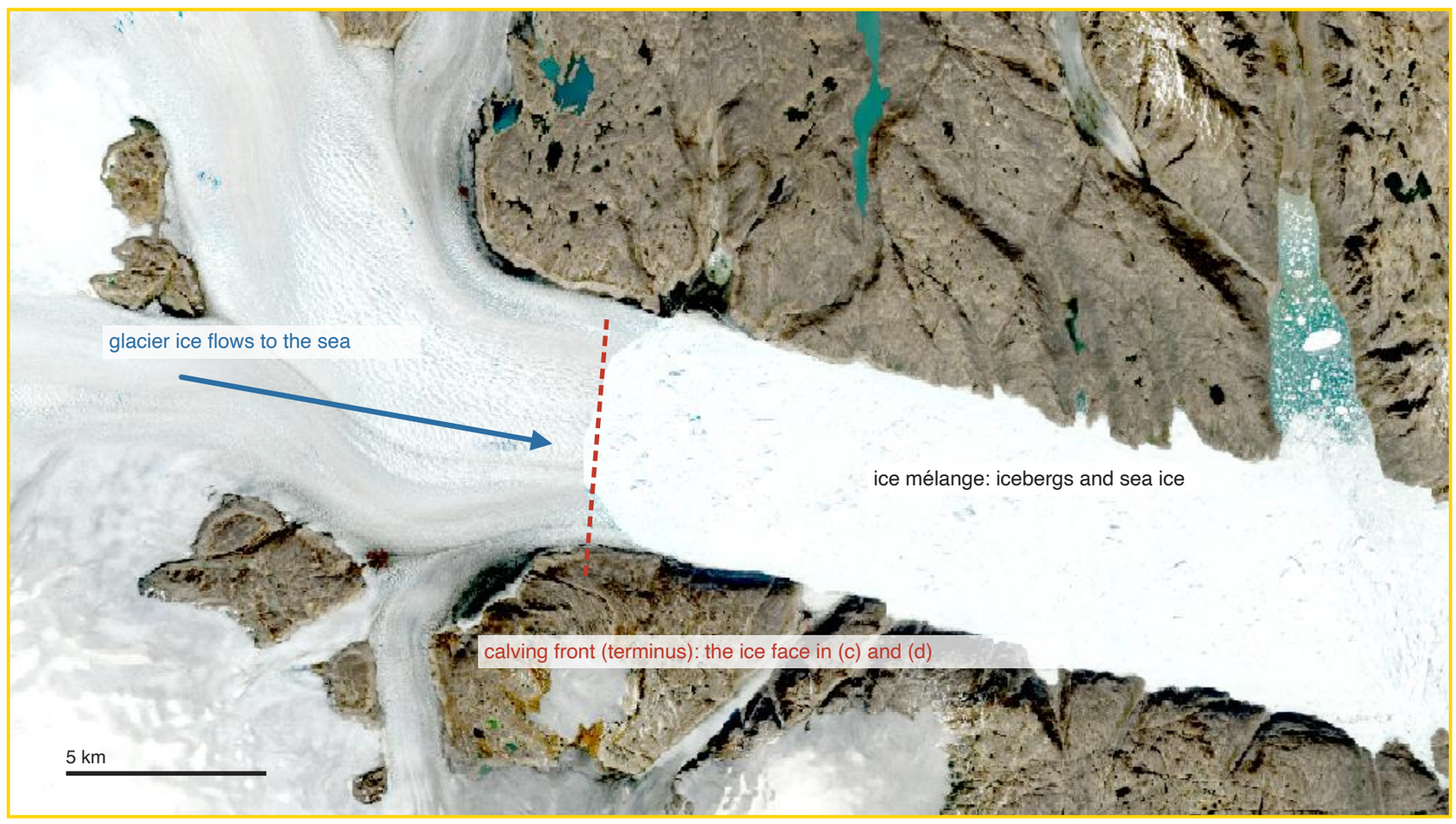
(a) Sermilik Fjord and Helheim Glacier, SE Greenland

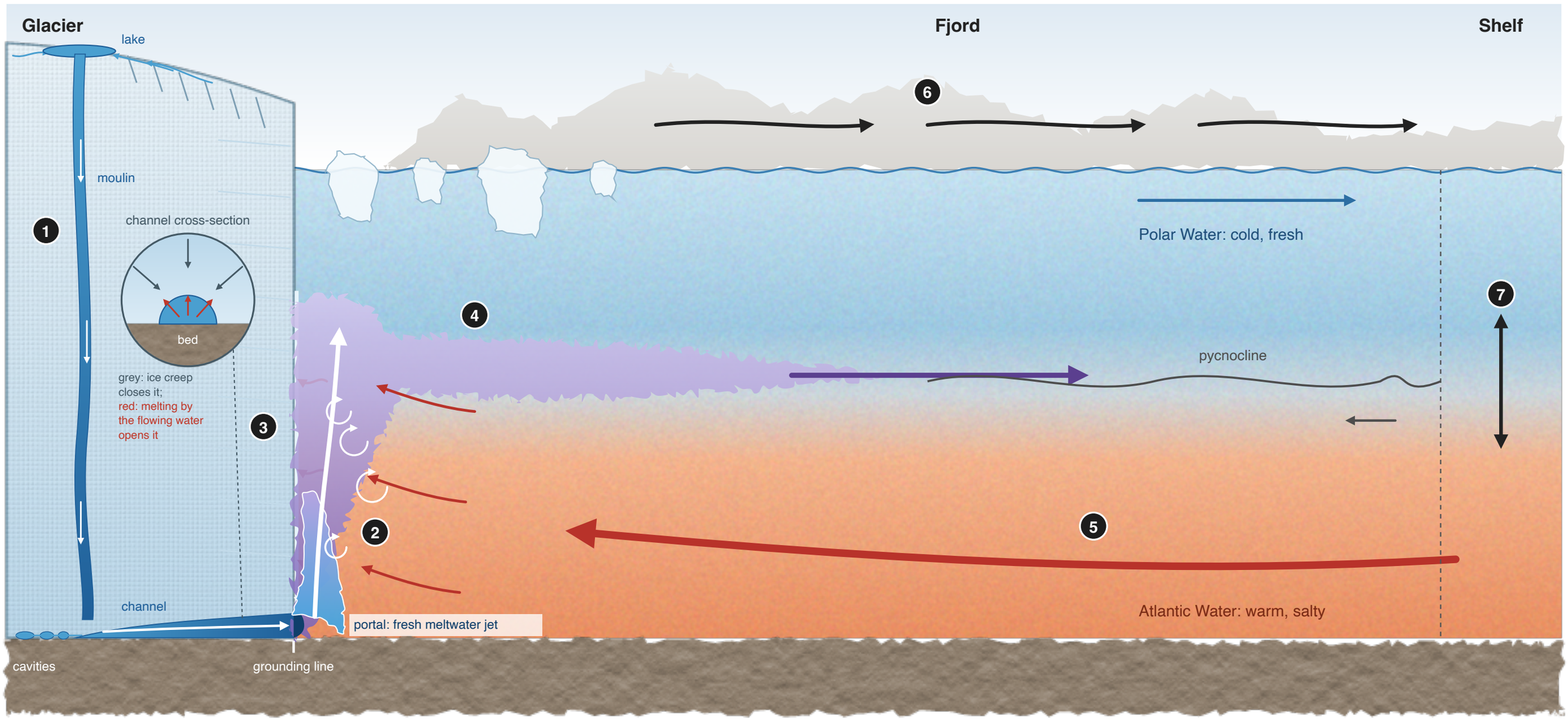


(b) Helheim Glacier terminus, 30 m resolution



Imagery: NASA Worldview / GIBS. (a) MODIS Terra true colour, 5 Aug 2022. (b) Harmonized Landsat Sentinel-2 (HLS S30), 30 Jul 2022; contains modified Copernicus Sentinel data 2022.

(c) Processes at a tidewater glacier and in its fjord

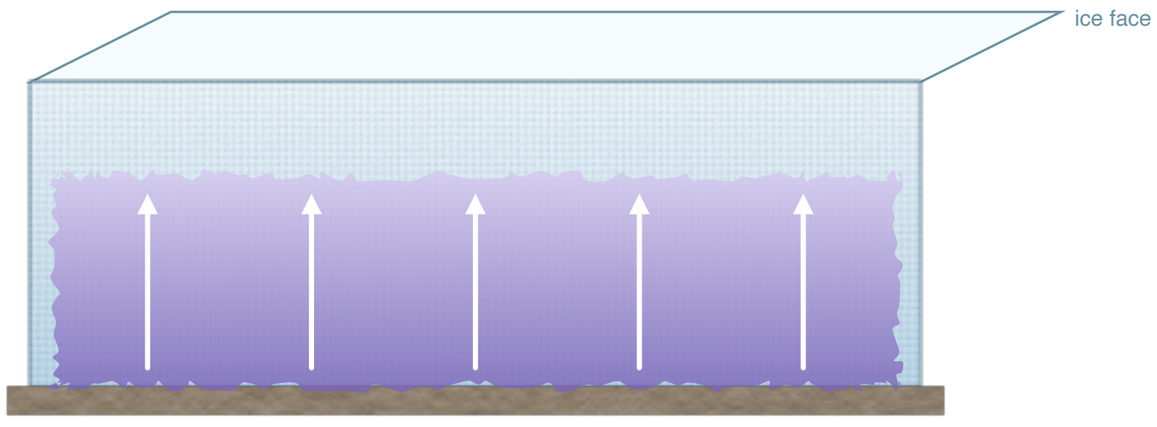


(c) is a schematic: not to scale; the plume and icebergs are enlarged.

1. subglacial discharge: meltwater drains via moulins to channels at the bed, exits at the grounding line
2. buoyant plume: rises along the ice and entrains ambient water (turbulent eddies)
3. submarine melting: the plume carries heat to the ice face
4. neutral buoyancy: the plume detaches and flows out toward the sea
5. warm Atlantic Water flows toward the glacier at depth, replacing the entrained water
6. along-fjord wind drives an exchange with the shelf
7. shelf forcing: the pycnocline heaves over days and the motion enters the fjord

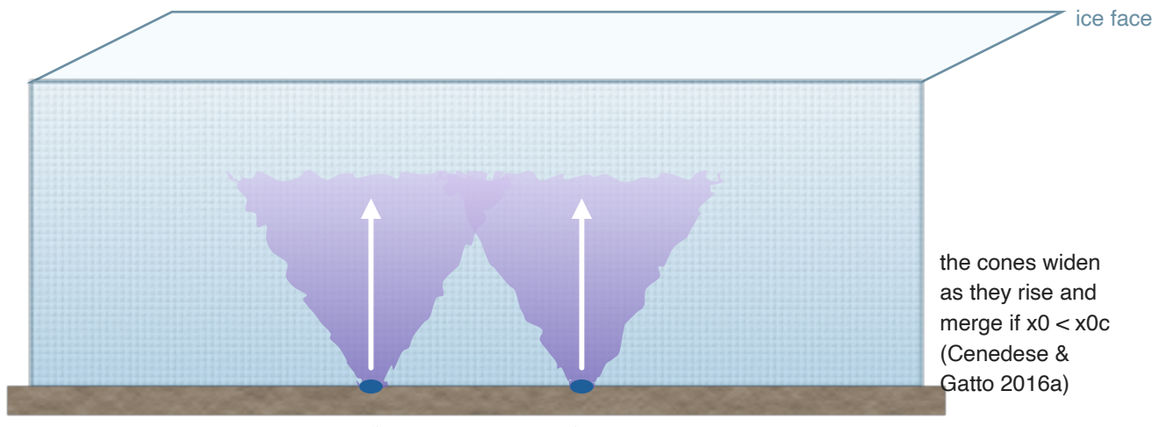
(d) Plume geometry at the ice face: configurations (a) and (b)

configuration (a): a line plume, discharge along the whole face



q0 per metre of ice face (2-D)

configuration (b): two point plumes, x0 apart



the cones widen as they rise and merge if x0 < x0c (Cenedese & Gatto 2016a)